

THE ERICE STATEMENT

*From the Work Session on Environmental Endocrine-Disrupting Chemicals:
Neural, Endocrine, and Behavioral Effects held in Erice, Sicily, Nov. 5-10, 1995*

THE PROBLEM

A multidisciplinary group of international experts gathered for a work session on "Environmental Endocrine-Disrupting Chemicals: Neural, Endocrine and Behavioral Effects" under the auspices of the International School of Ethology at the Ettore Majorana Centre for Scientific Culture in Erice, Sicily, November 5-10, 1995. The need for this work session grew out of evidence accumulated since the first consensus statement on endocrine-disrupting chemicals was released in October, 1991.

This first statement was framed as a product of a work session on "Chemically Induced Alterations in Sexual and Functional Development: The Wildlife/Human Connection." Research since 1991 has reinforced concerns over the scope of the problems posed to human health and ecological systems by endocrine-disrupting chemicals. New evidence is especially worrisome because it underscores the exquisite sensitivity of the developing nervous system to chemical perturbations that result in functional abnormalities. Moreover, the consequences of these perturbations depend upon the stage of development during which exposure occurs and are expressed in different ways at different times in life, from birth through to advanced age. This work session was convened because of the growing concern that failure to confront the problem could have major economic and societal implications. Those in attendance agreed that as scientists we seek only the truth; we value diversity; we believe global problems require global solutions; and our goal should be "science without borders and laboratories without walls" (adapted from: Paul Dirac, Piotr Kapitza, and Antonio Zichichi, Erice Statement, 1982).

The meeting was convened specifically to:

1. come to agreement in principle concerning the magnitude and geographic scope of the impact of endocrine disruptors on brain development and behavior;
2. review available technologies for ascertaining biologic markers of exposure to and effects on the nervous system by endocrine disruptors;
3. provide strategies for increasing communications and collaboration among disciplines to optimize resources for future research; and
4. suggest methods for translating the findings of this work session into information that is useful for decision-makers and the public.

CONSENSUS STATEMENT

The following consensus was reached by participants at the workshop.

1. We are certain of the following:

- Endocrine-disrupting chemicals can undermine neurological and behavioral development and subsequent potential of individuals exposed in the womb or, in fish, amphibians, reptiles, and birds, the egg. This loss of potential in humans and wildlife is expressed as behavioral and physical abnormalities. It may be expressed as reduced intellectual capacity and social adaptability, as impaired responsiveness to environmental demands, or in a variety of other functional guises. Widespread loss of this in nature can change the character of human societies or destabilize wildlife populations. Because profound economic and social consequences emerge from small shifts in functional potential at the population level, it is imperative to monitor levels of contaminants in humans, animals, and the environment that are associated with disruption of the nervous and endocrine systems and reduce their production and release.

- Because the endocrine system is sensitive to perturbation, it is a likely target for disturbance. In contrast to natural hormones found in animals and plants, some of the components and by-products of many manufactured

organic compounds that interfere with the endocrine system are persistent and undergo biomagnification in the food web, which makes them of greater concern as endocrine disruptors.

- Man-made endocrine-disrupting chemicals range across all continents and oceans. They are found in native populations from the Arctic to the tropics, and, because of their persistence in the body, can be passed from generation to generation. The seriousness of the problems is exacerbated by the extremely low levels of hormones produced naturally by the endocrine system which are needed to modulate and induce appropriate responses. In contrast, many endocrine-disrupting contaminants, even if less potent than the natural products, are presented in living tissue at concentrations millions of times higher than the natural hormones. Wildlife, laboratory animals, and humans exhibit adverse health effects at contemporary environmental concentrations of man-made chemicals that act as endocrine disruptors.

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New technology has revealed that some man-made chemicals are present in tissue at concentrations previously not possible to measure with conventional analytical methods, but at concentrations which are biologically active.

- Gestational exposure to persistent man-made chemicals reflects the lifetime of exposure of females before they become pregnant. Hence, the transfer of contaminants to the developing embryo and fetus during pregnancy and to the newborn during lactation is not simply a function of recent maternal exposure. For some egg-laying species, the body-burden of the females just prior to ovulation is the most critical period. For mammals, exposure to endocrine disruptors occurs during all of prenatal and early postnatal development because they are stored in the mother.

- The developing brain exhibits specific and often narrow windows during which exposure to endocrine disruptors can produce permanent changes in its structure and function. The timing of exposure is crucial during early developmental stages, particularly during fetal development when a fixed sequence of structural change is occurring and before protective mechanisms have developed. A variety of chemical challenges in humans and animals early in life can lead to profound and irreversible abnormalities in brain development at exposure levels that do not produce permanent effects in adults.

- Thyroid hormones are essential for normal brain functions throughout life. Interference with thyroid hormone function during development leads to abnormalities in brain and behavioral development. The eventual results of moderate to severe alterations of thyroid hormone concentrations, particularly during fetal life, are motor dysfunction of varying severity including cerebral palsy, mental retardation, learning disability, attention deficit hyperactivity disorder, hydrocephalus, seizures, and other permanent neurological abnormalities. Similarly, exposure to man-made chemicals during early development can impair motor function, spatial perception, learning, memory, auditory development, fine motor coordination, balance, and attentional processes; in severe cases, mental retardation may result.

- Sexual development of the brain is under the influence of estrogenic (female) and androgenic (male) hormones. Not all endocrine disruptors are estrogenic or anti-estrogenic. For example, new data reveal that DDE, a breakdown product of DDT, found in almost all living tissue, is an anti-androgen in mammals. Man-made chemicals that interfere with sex hormone share the potential to disturb normal brain sexual development. Wildlife studies of gulls, terns, fishes, whales, porpoises, alligators, and turtles link environmental contaminants with disturbances in sex hormone production and/or action. These effects have been associated with exposure to sewage and industrial effluents, pesticides, ambient ocean and freshwater contamination, and the aquatic food web.

- Commonalties across species in the hormonal mechanisms controlling brain development and function mean that adverse effects observed in wildlife and in laboratory animals may also occur in humans, although specific effects may differ from species to species. Most important, the same man-made chemicals that have shown these effects in mechanistic studies in laboratory animals also have a high exposure potential for humans.

- The full range of substances interfering with natural endocrine modulation of neural and behavioral development cannot be entirely defined at present. However, compounds shown to have endocrine effects include dioxins, PCBs, phenolics, phthalates, and many pesticides. Any compounds mimicking or antagonizing actions of, or altering levels of, neurotransmitters, hormones, and growth factors in the developing brain are potentially in this group.

2. We estimate with confidence that:

- Every pregnant woman in the world has endocrine disruptors in her body that are transferred to the fetus. She also has measurable concentrations of endocrine disruptors in her milk that are transferred to the infant.

- There may not be definable thresholds for responses to endocrine disruptors. In addition, for naturally occurring hormones, too much can be as severe a problem as too little. Consequently, simple (monotonic) dose-response curves for toxicity do not necessarily apply to the effects of endocrine disruptors.

- Because certain PCBs and dioxins are known to impair normal thyroid function, we suspect that they contribute to learning disabilities, including attention deficit hyperactivity disorder and perhaps other neurological abnormalities. In addition, many pesticides affect thyroid function and, therefore, may have similar consequences.

- Some endocrine disruptors or their breakdown products are nearly equipotent to natural hormones. Even weak endocrine disruptors may exert potent effects because they can bypass the natural protection of blood-binding proteins for endogenous hormones. Some disruptors also have a substantially longer biological half-life than naturally produced hormones because they are not readily metabolized, and as a result are stored in the body and accumulate to concentrations of concern. Some man-made chemicals that appear non-toxic are converted by the liver to more toxic compounds. Also, compounds that are not toxic in the mother may be toxic to her developing embryo, fetus or newborn. The exquisite vulnerability of the fetal brain to methylmercury and lead are prime examples of this principle.

- Functional deficits are not as easily measured as physical anomalies or clinical disease, in part because they are typically expressed as continuous measures, such as IQ, rather than the number of cases in a population. Consequently, conventional population surveys may overlook the extent of such deficits. Moreover, because such surveys tend to express their findings as shifts in mean values even when they are based on appropriate measures, they tend to obscure influences on the more susceptible members of the population.

- Large amounts of man-made chemicals capable of disrupting the endocrine and nervous systems are sold to, or produced and used in, Third World countries that lack the resources or technology to properly monitor and control exposure levels. Insufficient and improper training in handling chemicals and ignorance concerning health effects and monitoring strategies leads to the likelihood of very high levels of exposure.

3. There are many uncertainties in our understanding because:

- No one is exposure-free, thus confounding studies to determine what is normal. Everyone is exposed at any single time and throughout life to large numbers of manmade chemicals. Relatively few of the manmade chemicals found in human tissue have been identified. Lack of funding has seriously constrained testing these chemicals for their potential to disrupt natural systems.

- Sensitive parameters, including neurological abnormalities, behavioral and neuropsychiatric disorders, and neuroanatomical, neurochemical, and neurophysiologic endpoints need to be investigated. Most important, criteria at the population level need to include the social and economic costs of impairment because the true costs to society of such problems can be significant, for example, the costs of a 5 point IQ loss across a population. Investigation of potential toxicity typically includes laboratory, population and field studies, clinical reports, and accident reports. However, developmental neurotoxicants produce a spectrum of effects that are not typically evaluated, such as the

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progression and latency of behavioral and neurological changes. In addition, alteration of other systems can produce subsequent cognitive, behavioral, and neurological dysfunction; that is, diseases of other organ systems that influence the brain; non-CNS drugs; other foreign substances such as air pollutants; and immune system involvements that alter behavior.

- Trade secret laws afford industry confidentiality, depriving the consumer and public health authorities of the right to know the components of commercial products so they can be tested.

4. Our judgment is that:

- The benefits of reduced health care costs could be substantial if exposures to endocrine-disrupting chemicals were reduced.

- A trivial amount of governmental resources is devoted to monitoring environmental chemicals and health effects. The public is unaware of this and believes that they are adequately protected. The message that endocrine disruptors are present in the environment and have the potential to affect many people over a lifespan has not effectively reached the general public, the scientific community, regulators, or policy makers. Although this message is difficult to reduce to simple statements without over- or understating the problem, the potential risks to human health are so widespread and far-reaching that any policy based on continued ignorance of the facts would be unconscionable.

- The outcome of exposure is inadequately addressed when based just on population averages. Instead, risk should be based on the range of responses within a population — that is, the total distribution. The magnitude of the problem can be better determined by knowing the distribution of responses to endocrine disruptors by individuals within subsets of the population most at risk, such as pregnant women, developing embryos, fetuses, and newborns, teens, the aged, the ill or those with pre-existing endocrine disorders. The magnitude of the risks also depends upon the endpoint under consideration. For example, a variety of motor, sensory, behavioral, and cognitive functions, endpoints which are more sensitive than cancer, must be considered when assessing neurological function. This holds for wildlife and domestic animals, as well as human populations.

- Wildlife have been effective models for understanding endocrine disruption at the molecular, cellular, individual, population, and ecosystem levels. Future research to examine diverse wildlife species at all levels of biological organization must be broadened and adequately supported.

- Those responsible for producing man-made chemicals must assure product safety beyond a reasonable doubt. Manufacturers should be required to release the names of all chemicals used in their products with the appropriate evidence that the products pose no developmental health hazard.

- Current panels of scientists who determine the distribution of public research funds often have a narrow scope of expertise and are thus ill-equipped to review the kind of interdisciplinary research that is necessary in this field. Funding institutions should be encouraged to increase the scope of representation on review panels and to develop more appropriate mechanisms for interdisciplinary reviews. Governmental agencies should also increase funding for multidisciplinary extramural projects for surveillance of wildlife and human populations where neurological damage is suspected and follow any leads with laboratory research. In addition, populations of animals consuming the contaminated foods also eaten by humans should be studied for developmental health effects. It is important to observe a variety of vertebrate species through multigenerational studies.

- Strategies for increasing interdisciplinary communication and collaborations to optimize resources and future research are needed. Studies should be designed more economically to include the sharing of material among many collaborators. Interdisciplinary teams should explore neurological and other types of damage at all levels of biological organization from molecular through biochemical, physiological, and behavioral.

- A concerted effort should be undertaken to deliver this consensus statement to the public, key decision-makers, and the media. In addition, specially designed messages should be developed for family physicians and others responsible for public health who are often unaware of the possible role of occupational and environmental chemical pollutants as agents underlying or constituting risk factors for "primary" human diseases. Physicians must be trained in medical school about often latent effects of pollutants on human development and health. This training is currently inadequate. A coordinated speakers bureau and on-line systems such as a site on the World Wide Web for endocrine-disruptors should be established.

The scientists listed below signed the statement. They were all participants in the meeting in Erice in November 1995 when and where the statement was written.

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